

MINGYU SEONG

Wearable Robotics | Lower-Limb Exoskeletons | Rehabilitation Robotics | Human-Robot Interaction
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EDUCATION

Yeungnam University, Department of Robotics Engineering M.S. in Robotics Engineering GPA: 4.37 / 4.50 <i>Thesis: "Development of Personalized Robotic Achilles Tendon based on Switchable-Stiffness Spring"</i> Advisor: Prof. Jungsu Choi Relevant Coursework (all A+): Advanced Automatic Control, Robot Dynamics and Control, Automotive Control Engineering, Advanced Motor Design, Advanced Linear Algebra, Advanced Robotics	Feb. 2026
Yeungnam University, Department of Robotics Engineering B.Eng. in Robotics Engineering GPA: 4.20 / 4.50	Feb. 2024

RESEARCH EXPERIENCE

Human Robot Convergence Laboratory (HRC Lab), Yeungnam University <i>Researcher Mar. 2021 – Present Advisor: Prof. Jungsu Choi</i> - Developed a 0.99-kg semi-passive ankle exoskeleton using switchable-stiffness pneumatic springs (M.S. thesis; patent pending). - Led the full development cycle: mechanism design, sensing and gait-phase control, and human-subject testing. - Reduced the metabolic cost of walking by 14.3% compared with walking without the device.	Gyeongsan, Korea
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PUBLICATIONS

<i>Journal Article (Under Review)</i> M. Seong and J. Choi*, "Robotic Achilles Tendon: A Lightweight Semi-Passive Ankle-Assist System for Reducing the Metabolic Cost of Walking," <i>IEEE Robotics & Automation Magazine</i> , under review.	
<i>Peer-Reviewed International Conference Papers</i> M. Seong , H. Heo, H. Lee, and J. Choi*, "Personalized Robotic Achilles Tendon Utilizing a Semi-Passive Spring with Switching Stiffness," <i>IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i> , 2025. DOI: 10.1109/IROS60139.2025.11245843	
M. Seong , H. Heo, H. Lee, and J. Choi*, "A Novel Wearable Ankle Assistive System Utilizing Switching Stiffness Pneumatic Spring," <i>International Conference on Control, Automation, and Systems (ICCAS)</i> , 2025. DOI: 10.23919/ICCAS66577.2025.11301378	
<i>Domestic Conference Papers (in Korean, with English abstracts)</i> M. Seong , H. Heo, and J. Choi*, "Development of Bioinspired Variable Stiffness Actuating System based on Human Musculoskeletal Characteristics," <i>ICROS Annual Conference</i> , 2025.	
M. Seong , H. Heo, and J. Choi*, "Development of Semi-passive Ankle Assist System Based on Lower Limb Dynamic Characteristic," <i>Korea Robotics Society Annual Conference (KRoC)</i> , 2025.	
M. Seong and J. Choi*, "Development of Bioinspired Force Transmission Structure based on Human Sarcomere by using Fluid Dynamics," <i>ICROS Annual Conference</i> , 2024. — Outstanding Paper Award	
M. Seong , Y. Lee, and J. Choi*, "New Measurement Approach for Dynamic Force Utilizing Fluid," <i>Korea Robotics Society Annual Conference (KRoC)</i> , 2023.	
* Corresponding author	

CONFERENCE PRESENTATIONS

M. Seong , H. Heo, and J. Choi*, "Semi-passive Ankle Assist System Based on Lower Limb Dynamic Characteristics of Human Walking," Oral Presentation, <i>Joint 10th IFAC Symposium on Mechatronic Systems and 14th IFAC Symposium on Robotics</i> , Paris, France, July 2025.

PATENTS

J. Choi and M. Seong , "Ankle Assist Device," Korean Patent Application No. 10-2026-0011170, filed Jan. 20, 2026 (pending).
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SELECTED FUNDED RESEARCH PROJECTS (PARTICIPATING RESEARCHER)

Inner-Wear-Type Wearable Parasitic Muscle Suit to Overcome the Limitations of Human Physical Ability National Research Foundation of Korea (NRF) — Outstanding Young Scientist Grants	Mar. 2023 – Feb. 2028
HD-DNA Lab: Human Behavior Stem Information-Based Digital Human Cloning Innovative Supplementary Engineering Laboratory National Research Foundation of Korea (NRF) — Basic Research Laboratory (Pioneer)	Jun. 2022 – Feb. 2025
Develop of Exomuscle Device to Improve a Gait Disturbance for People with Depressed Ankle Motion National Research Foundation of Korea (NRF) — Basic Research Grant	Jun. 2021 – Feb. 2023
Development of High-Speed Winding Equipment for Tow Prepreg-Based Hydrogen Storage Vessels Korea Evaluation Institute of Industrial Technology (KEIT) — Materials and Components Technology Development Program	Jun. 2021 – Jun. 2023

HONORS & AWARDS

The Presidential Science Scholarship — MSIT / Korea Student Aid Foundation (KOSAF) <i>Nationwide selection of 60 third-year undergraduates (2022); full tuition through graduation plus academic stipend. Also selected for the National Scholarship for Science and Engineering (2022), superseded under non-duplication rules.</i>	2022 – 2023
Outstanding Paper Award — ICROS 2024, Institute of Control, Robotics and Systems	2024

Gold Medal — YU Capstone Design Project Competition, Yeungnam University	2023
Silver Medal — Autonomous Driving Competition, TE Connectivity	2023
Special Award — Autonomous Driving Competition, Yeungnam University	2023
Bronze Medal — Extreme Environmental Drone Competition, Yeungnam University	2023
Silver Medal — Extreme Environmental Drone Competition, Yeungnam University	2022
Grand Award — Self-Following Mobile Robot Competition, Yeungnam University	2021
Silver Medal — Y-Star Startup Item Competition, Yeungnam University	2021

TECHNICAL SKILLS

Software: MATLAB, Python, LabVIEW, SolidWorks, AutoCAD

Hardware & Experiment: Arduino, NI myRIO, 3D Printing, EMG, Metabolic Measurement

LEADERSHIP & SERVICE

President, Yeungnam University Robot Club (Power Supply)	Jan. – Dec. 2023
Republic of Korea Army — Engineer (Military Service)	Apr. 2019 – Nov. 2020